WEEK 10: ASSIGNMENT

1. For the table shown below, match the following correctly:

Voltage at the CP line	State
I. +12V	a. A
II. +6V	b. B
III. +9V	c. C
IV. +3V	d. D
V. -12V	e. E
VI. 0V	f. F

A. I-a, II-b, III-c, IV-d, V-e, VI-f
B. I-a, II-d, III-c, IV-b, V-f, VI-e
B. I-b, II-c, III-d, IV-a, V-e, VI-f
D. I-a, II-c, III-b, IV-d, V-f, VI-e

Answer: D

2. If the maximum current which the EV can demand as per IEC 61851-1 in case of AC type-2 charging is 36 A, then what will be the duty ratio (in percentage) of PWM signal.

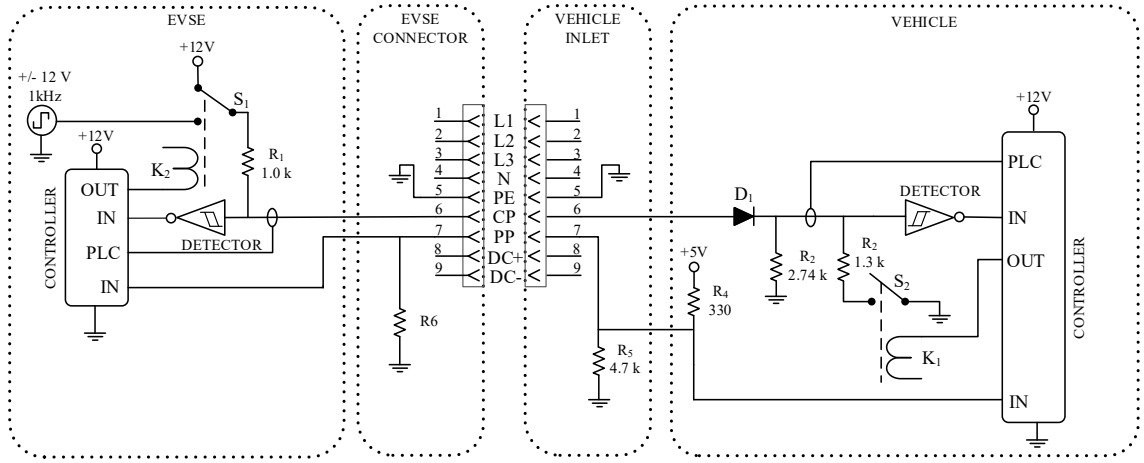
Answer: 60

3. If the maximum current which the EV can demand as per IEC 61851-1 in case of AC type-2 charging is 48 A, then what will be the time (in μs) during which positive voltage level will be applied on the CP line in one PWM cycle.

Answer: 800

Range: 790 to 810

4. For the AC connector, below is the equivalent circuit

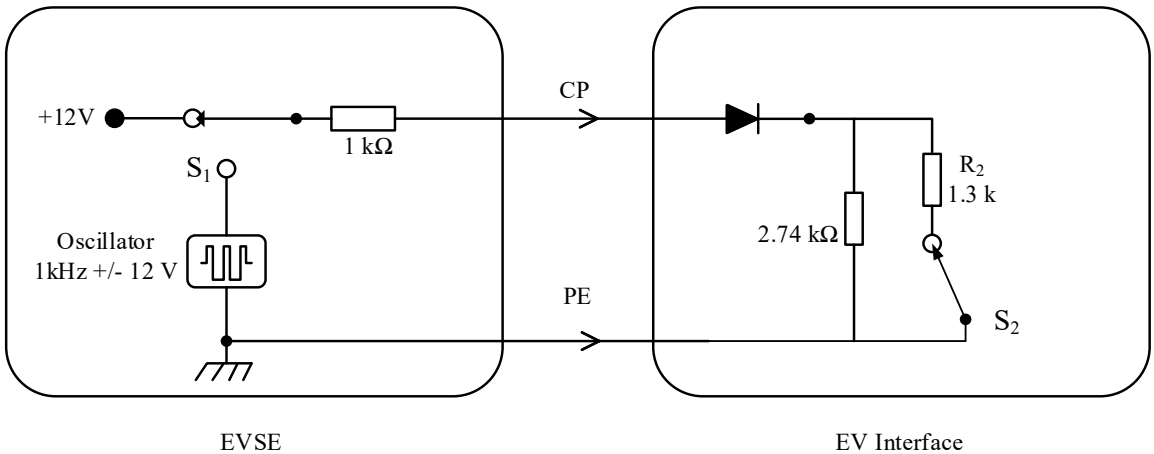


When it is being plugged in the vehicle inlet, what will be the voltage at PP pin (round it to two decimal points), if $R_6 = 300\Omega$.

Answer: 2.30

Range: 2.1 to 2.7

5. For the CCS2 charging, if at any given point of instant, the equivalent circuit of CP signaling is shown below, then what will be the voltage at CP pin with respect to PE (round it to two decimal points).



Answer: 5.62

Range: 5 to 6.5

6. Which of the following statement/statements are true for air-gap in the inductors:
(A) Air-gap in inductor allows larger Ampere-turns for which core enters the saturation
(B) Air-gap in inductor increases the inductance value
(C) Air-gap in inductor reduces the saturation flux density of the core material
(D) Air-gap in inductor makes the inductance value independent on the core dimensions

Answer: A, D

7. Which of the following statement/statements are true for SAE J1772 signaling protocol:
A. The voltage level on the CP line is detected by the EV side controller
B. The voltage level on the CP line is detected by the EVSE side controller
C. The duty ratio and frequency on the CP line is detected by the EV side controller
D. The duty ratio and frequency on the CP line is detected by the EVSE side controller

Answer: B, C

8. Which of the following statement/statements are true for AC Type-2 charging:
A. CP is responsible for detecting the insertion of correct plug
B. PP is used to communicate the status of charging
C. PP is responsible for detecting the insertion of correct plug
D. CP is responsible to communicate the status of charging

Answer: C, D

9. Which of the following statement/statements is/are true for CCS-2 DC charging:
A. In CCS2, the power line communication takes place over CP-PE pin
B. In CCS2, the power line communication takes place over DC+ - DC- pin
C. In CCS2, the power line communication takes place over PP - PE- pin

Answer: A

10. What possible duty ratio (in percentage) could be there on CP pin when AC Type-2 charger plug is plugged in CCS2 vehicle inlet connector.

A. 30
B. 5
C. 60
D. 4

Answer: A,C

Answer Key:

1. D	2. 60
3. 790 to 810	4. 2.1 to 2.7
5. 5 to 6.5	6. A, D
7. B, C	8. C, D
9. A	10. A, C